

Math 140 Spring 2023 - Worksheet 4.6

Section:

Name:

1. An outdoor track is to be created in the shape of a rectangle with semicircles at two opposite ends. If the perimeter of the track is 440 yards, find the dimensions of the track for which the area of rectangular portion is maximized.
2. A rectangular box open at the top has vertical sides, a "square" bottom, and a volume of 8000 cubic meters. Find its dimensions having the least possible surface area. (Hint: First, write the surface area as a function of one variable only.)
3. At 7 PM, an oil tanker traveling west at 18 km/h passes the same spot as a luxury liner that arrived at the same spot at 6 PM while traveling north at 36 km/h. Setting this spot as the origin, find the location of the oil tanker and the luxury liner t hours after 6 PM. Then find the distance between the two as a function of time. At what time were the ships closest to each other?
4. Hyunah bakes tiramisu at the cost of $C(x) = (x - 10)^2 + 20$ in dollars when baking x many units, and sells at \$100 per unit. Find how many tiramisu Hyunah should sell to maximize the profit.